



**MANIPAL UNIVERSITY
JAIPUR**

(University under Section 2(f) of the UGC Act)



B.TECH SECOND YEAR

ACADEMIC YEAR: 2024-25



COURSE NAME: STATISTICAL METHODS AND PROBABILITY THEORY

COURSE CODE : MAS 2001

LECTURE SERIES NO :

CREDITS : 03

MODE OF DELIVERY :

FACULTY :

EMAIL-ID :

PROPOSED DATE OF DELIVERY:

ASSESSMENT PLAN

Criteria	Description	Max. Marks
Internal Assessment (Summative)	Sessional Exam	30
	Quizzes and Assignments	30
End Term Exam (Summative)	End Term Exam	40
	Total	100

SYLLABUS

Probability Theory and Random Variables: Probability (Only One Lecture), Random variables, Cumulative distribution functions, Discrete random variables, Continuous random variables, Independent random variables, Probability mass and density functions, Expectation of random variables, Chebyshev's inequality, Central limit theorem. Probability distribution: Binomial, Poisson, Uniform, Normal, Exponential Theory of Estimation: Maximum Likelihood and method moment estimation, Sufficient statistics, Bayesian estimation, Confidence intervals for means. Tests of Statistical Hypothesis: Introduction, Parameter and Statistic, Standard error, Statistical hypotheses, Critical region, Tests of hypotheses and significance, Type I and Type II errors, level of significance. Test about one mean, Test about equality of two means, Test of variances, Chi square test, Analysis of Variance.

REFERENCES BOOKS

1. Goon, A.M., Gupta, M.K. and Dasgupta, B., An Outline of Statistical Theory, Vol. II, 3rd Edn. World Press, Kolkata, 2005.
2. Kendall and Stuart, Advanced Theory of Statistics Vol.-II, Charles Griffin & Co. Ltd, London, 1961.
3. Casella, G. and Berger, R.L., Statistical Inference, Second Edn. Thomson Duxbury, 2002.
4. Hogg, R.V. and Tanis, E.A., Probability and Statistical Inference, 3rd Edn. Macmillan Publishing Co. Inc., 1988.
5. Feller, W., An Introduction to Probability Theory and Its Applications, Vol. 1, 3rd Edition, John Wiley, 1968.

We will usually denote a set by a capital letter.

If x is an element in A , we write $x \in A$. Otherwise, we write $x \notin A$.

Some commonly used sets

$\mathbb{N}$: the natural numbers, $\{1, 2, 3, \dots\}$

$\mathbb{Z}$: the integers, $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

$\mathbb{Q}$: the rational numbers, $\{a/b \mid a, b \in \mathbb{Z}, b \text{ is not equal to } 0\}$.

$\mathbb{R}$: the real numbers

$\mathbb{C}$: the complex numbers, $\{a + bi \mid a, b \in \mathbb{R}\}$, where $i^2 = -1$.

The vertical line, $|$, means “such that”, or “where”. We can also use a colon for this. Commas are read as “and”.

There are often multiple ways to describe a set, e.g.,

$$\{x \in \mathbb{R} \mid x^2 - 5x + 6 = 0\} = \{x \mid x \in \mathbb{R}, x^2 - 5x = -6\} = \{2, 3\}.$$

A set is **finite** if it has a finite number of elements. Otherwise, it is an **infinite set**.

The number of elements in a set A is called its **cardinality**, denoted $|A|$. If A is infinite, we may write $|A| = \infty$.

Definition

Let A and B be sets. We say that A is a **subset** of B if (and only if) every element of A is an element of B . We write this as $A \subseteq B$, or $B \supseteq A$.

BASIC SET OPERATIONS

Definition

The **intersection** of sets A and B is the set of elements in both A and B , denoted

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

Two sets are **disjoint** if they have no elements in common, i.e., if $A \cap B = \emptyset$. The **union** of sets A and B is the set of elements in *either* A or B , denoted $A \cup B = \{x \mid x \in A \text{ or } x \in B \text{ or both}\}$.

Examples

1. If $A = \{2, 5, 8\}$ and $B = \{7, 5, 22\}$, then $A \cap B = \{5\}$ and $A \cup B = \{2, 5, 8, 7, 22\}$.
2. $Z \cup Q = Q$, and $Z \cap Q = Z$.
3. $A \cup \emptyset = A$ for any set A .

SET COMPLEMENTS

Frequently, we will need to establish the set of all elements U under consideration, which we call the **universal set**.

Definition

The **complement** of a set A is the set of all elements in U that are not in A :

$$A^c = \{x \in U \mid x \notin A\}.$$

RELATIVE COMPLEMENTS

Definition

For sets A and B , the **complement of A relative to B** is the set of elements that are in B but not A :

$$B - A = \{x \mid x \in B \text{ and } x \notin A\}.$$

The **symmetric difference** of A and B is the set of elements that are in one of these sets, but not the other:

$$A \oplus B = (A - B) \cup (B - A).$$

The complement of A relative to B can be denoted $A \setminus B$.

Exercises

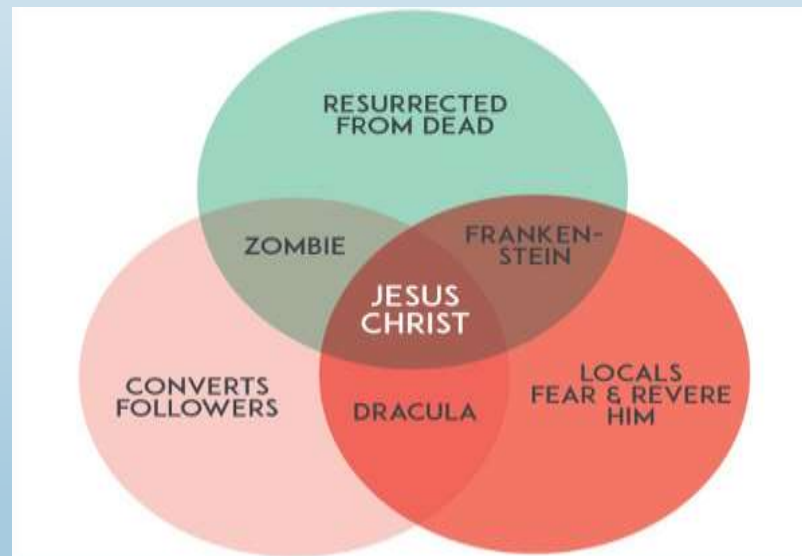
Compute $A - B$, $B - A$, and $A \oplus B$ in the following cases:

1. $A = \{1, 3, 8\}$ and $B = \{2, 4, 8\}$
2. Any set A , and $B = \emptyset$.
3. $A = \mathbb{R}$ and $B = \mathbb{Q}$.

Venn diagrams

A useful way to visualize a small number of sets and their intersections, unions, and relative complements, is with a **Venn diagram**.

Social media has caused these to become mainstream, though they are often used incorrectly.



CARTESIAN PRODUCTS

Definition

The **Cartesian product** of sets A and B is the set of ordered pairs:

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

Examples

Let $A = \{1, 2, 3\}$ and $B = \{4, 5\}$. Then

$$A \times B = B \times A = A \times A =$$

Similarly, we can define the Cartesian product of three (or more) sets. For example,

$$A \times B \times C = \{(a, b, c) \mid a \in A, b \in B, c \in C\}.$$

It is common to use exponents if the sets are the same, e.g.,

$$A^2 = A \times A, \quad A^3 = A \times A \times A, \dots$$

POWER SETS

Definition

The **power set** of A is the set of all subsets of A , denoted $P(A)$. (Including both $\emptyset$ and A .)

Examples

1. $P(\emptyset) = \{\emptyset\}$
2. $P(\{1\}) = \{\emptyset, \{1\}\}$
3. $P(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$

SUMMATION NOTATION

Addition is a **binary operation** that is **associative**, which means that parentheses are **permitted anywhere but required nowhere**.

As such, we may write

$$((a_1 + a_2) + a_3) + a_4 = (a_1 + a_2) + (a_3 + a_4) = a_1 + a_2 + a_3 + a_4 = \sum_{k=1}^4 a_k,$$

and the last term is called **summation notation**.

A **finite series** is an expression such as $a_1 + a_2 + \cdots + a_n = \sum_{k=1}^n a_k$. We say:

- the variable k is the **index**
- the expression a_k is the **general term** of the series

the values below and above the summation symbol are the **initial index** and **terminal index**, respectively.

Another associative binary operation is **multiplication**. The product of elements $a_1, \dots, a_n$ is written in **product notation**, using a $\prod$ instead of a Σ :

$$a_1 a_2 \cdots a_n = \prod_{k=1}^n a_k.$$

ASSOCIATIVE SET OPERATIONS

Let $A_1, A_2, \dots, A_n$ be sets, then

$$(a) A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$$

$$(b) A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$$

$$(c) A_1 \times A_2 \times \dots \times A_n = \prod_{i=1}^n A_i$$

$$(d) A_1 \oplus A_2 \oplus \dots \oplus A_n = \bigoplus_{i=1}^n A_i.$$

DISTRIBUTIVE LAWS

See if you can find a general formula for the following two expressions by looking at the cases where $n = 2$ and drawing a Venn diagram:

$$\mathbf{A} \cap \left(\bigcup_{i=1}^n B_i \right) =$$

$$\mathbf{A} \cup \left(\bigcap_{i=1}^n B_i \right) =$$

PROBABILITY THEORY

Basics

Probability theory deals with the study of random phenomena, which under repeated experiments yield different outcomes that have certain underlying patterns about them. The notion of an experiment assumes a set of repeatable conditions that allow any number of identical repetitions. When an experiment is performed under these conditions, certain elementary events occur in different but *completely uncertain* ways. We can assign nonnegative number as the probability of the event in various ways:

Laplace's Classical Definition: The Probability of an event A is defined a-priori without actual experimentation as

$$P(A) = \frac{\text{Number of outcomes favorable to } A}{\text{Total number of possible outcomes}}, \quad (1-1)$$

provided all these outcomes are *equally likely*.

Consider a box with n white and m red balls. In this case, there are two elementary outcomes: white ball or red ball. Probability of “selecting a white ball” $= \frac{n}{n+m}$.

We can use above classical definition to determine the probability that a given number is divisible by a prime p .

If p is a prime number, then every p^{th} number (starting with p) is divisible by p . Thus among p consecutive integers there is one favorable outcome, and hence

$$P\{a \text{ given number is divisible by a prime } p\} = \frac{1}{p} \quad (1-2)$$

Relative Frequency Definition: The probability of an event A is defined as

$$P(A) = \lim_{n \rightarrow \infty} \frac{n_A}{n} \quad (1-3)$$

where n_A is the number of occurrences of A and n is the total number of trials.

We can use the relative frequency definition to derive (1-2) as well. To do this we argue that among the integers

$1, 2, 3, \dots, n$, the numbers $p, 2p, \dots$ are divisible by p .

Thus there are n/p such numbers between 1 and n . Hence

$$P\{a \text{ given number } N \text{ is divisible by a prime } p\} \\ = \lim_{n \rightarrow \infty} \frac{n/p}{n} = \frac{1}{p}. \quad (1-4)$$

In a similar manner, it follows that

$$P\{p^2 \text{ divides any given number } N\} = \frac{1}{p^2} \quad (1-5)$$

and

$$P\{pq \text{ divides any given number } N\} = \frac{1}{pq}. \quad (1-6)$$

The axiomatic approach to probability, due to Kolmogorov, developed through a set of axioms (below) is generally recognized as superior to the above definitions, (1-1) and (1-3), as it provides a solid foundation for complicated applications.

The totality of all ξ_i , *known a priori*, constitutes a set Ω , the set of all experimental outcomes.

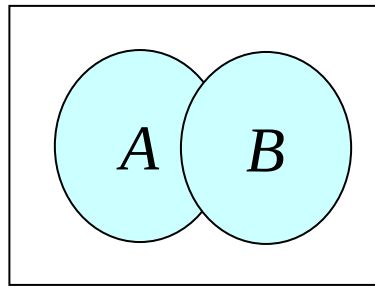
$$\Omega = \{ \xi_1, \xi_2, \dots, \xi_k, \dots \} \quad (1-7)$$

Ω has subsets $A, B, C, \dots$. Recall that if A is a subset of Ω , then $\xi \in A$ implies $\xi \in \Omega$. From A and B , we can generate other related subsets $A \cup B, A \cap B, \bar{A}, \bar{B}$, etc.

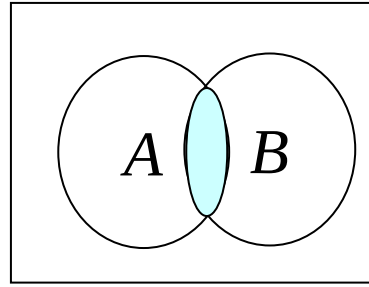
$$A \cup B = \{ \xi \mid \xi \in A \text{ or } \xi \in B \}$$
$$A \cap B = \{ \xi \mid \xi \in A \text{ and } \xi \in B \}$$

and

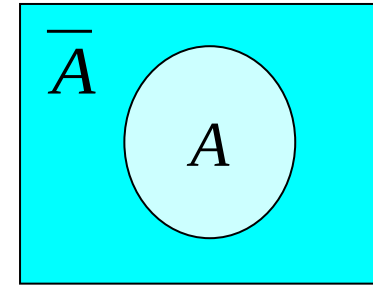
$$\bar{A} = \{ \xi \mid \xi \notin A \} \quad (1-8)$$



$$A \cup B$$



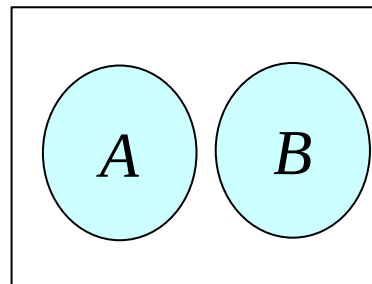
$$A \cap B$$



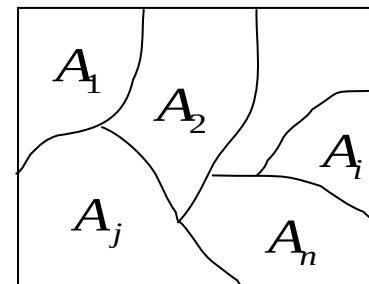
$$\bar{A}$$

- If $A \cap B = \phi$, the empty set, then A and B are said to be mutually exclusive (M.E).
- A partition of Ω is a collection of mutually exclusive subsets of Ω such that their union is Ω .

$$A_i \cap A_j = \phi, \text{ and } \bigcup_{i=1} A_i = \Omega. \quad (1-9)$$

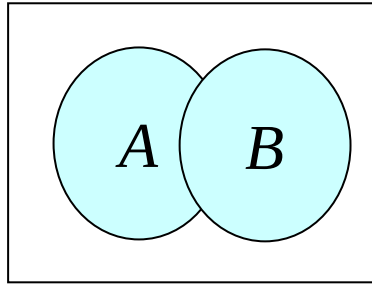


$$A \cap B = \phi$$

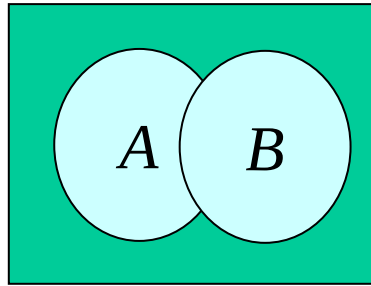


De-Morgan's Laws:

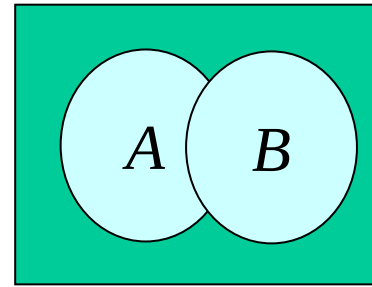
$$\overline{A \cup B} = \bar{A} \cap \bar{B}; \quad \overline{A \cap B} = \bar{A} \cup \bar{B} \quad (1-10)$$



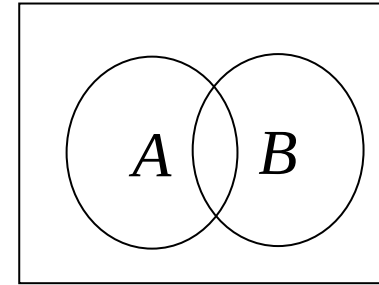
$A \cup B$



$\overline{A \cup B}$



$\bar{A} \cap \bar{B}$



- Often it is meaningful to talk about at least some of the subsets of Ω as events, for which we must have mechanism to compute their probabilities.

Example 1.1: Consider the experiment where two coins are simultaneously tossed. The various elementary events are

$$\xi_1 = (H, H), \quad \xi_2 = (H, T), \quad \xi_3 = (T, H), \quad \xi_4 = (T, T)$$

and

$$\Omega = \{ \xi_1, \xi_2, \xi_3, \xi_4 \}.$$

The subset $A = \{ \xi_1, \xi_2, \xi_3 \}$ is the same as “Head has occurred at least once” and qualifies as an event.

Suppose two subsets A and B are both events, then consider

“Does an outcome belong to A or B ” $= A \cup B$

“Does an outcome belong to A and B ” $= A \cap B$

“Does an outcome fall outside A ”?

Axioms of Probability

For any event A , we assign a number $P(A)$, called the probability of the event A . This number satisfies the following three conditions that act the axioms of probability.

- (i) $P(A) \geq 0$ (Probability is a nonnegative number)
 - (ii) $P(\Omega) = 1$ (Probability of the whole set is unity)
 - (iii) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$.
- (1-12)

(Note that (iii) states that if A and B are mutually exclusive (M.E.) events, the probability of their union is the sum of their probabilities.)

The following conclusions follow from these axioms:

a. Since $A \cup \bar{A} = \Omega$, we have using (ii)

$$P(A \cup \bar{A}) = P(\Omega) = 1.$$

But $A \cap \bar{A} \in \phi$, and using (iii),

$$P(A \cup \bar{A}) = P(A) + P(\bar{A}) = 1 \quad \text{or} \quad P(\bar{A}) = 1 - P(A). \quad (1-13)$$

b. Similarly, for any A , $A \cap \{\phi\} = \{\phi\}$.

Hence it follows that $P(A \cup \{\phi\}) = P(A) + P(\phi)$.

But $A \cup \{\phi\} = A$, and thus $P\{\phi\} = 0$. (1-14)

c. Suppose A and B are *not* mutually exclusive (M.E.)?

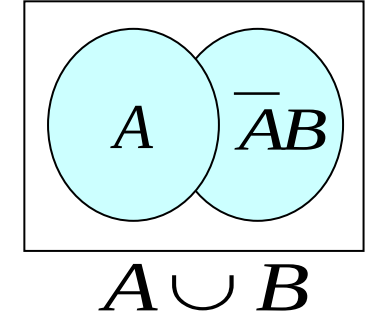
How does one compute $P(A \cup B) = ?$

To compute the above probability, we should re-express $A \cup B$ in terms of M.E. sets so that we can make use of the probability axioms. From Fig. we have

$$A \cup B = A \cup \bar{A}B, \quad (1-15)$$

where A and $\bar{A}B$ are clearly M.E. events.

Thus using axiom (1-12-iii)



$$P(A \cup B) = P(A \cup \bar{A}B) = P(A) + P(\bar{A}B).$$

$$B = B \cap \Omega = B \cap (A \cup \bar{A}) \quad (1-16)$$

$$= (B \cap A) \cup (B \cap \bar{A}) = BA \cup B\bar{A} \quad (1-17)$$

To compute $P(\bar{A}B)$, we can express B as

Thus

$$P(B) = P(BA) + P(B\bar{A}), \quad (1-18)$$

$BA = AB$ since $B\bar{A} = \bar{A}B$ and are M.E. events.

$$P(\bar{A}B) = P(B) - P(AB)$$

(1-19)

From (1-19), $P(A \cup B) = P(A) + P(B) - P(AB).$

(1-20)

and using (1-19) in (1-16)

Conditional Probability and Independence

In N independent trials, suppose N_A , N_B , N_{AB} denote the number of times events A , B and AB occur respectively. According to the frequency interpretation of probability, for large N

$$P(A) \approx \frac{N_A}{N}, \quad P(B) \approx \frac{N_B}{N}, \quad P(AB) \approx \frac{N_{AB}}{N}. \quad (1-21)$$

Among the N_A occurrences of A , only N_{AB} of them are also found among the N_B occurrences of B . Thus the ratio

$$\frac{N_{AB}}{N_B} = \frac{N_{AB} / N}{N_B / N} = \frac{P(AB)}{P(B)} \quad (1-22)$$

is a measure of “the event A given that B has already occurred”. We denote this conditional probability by

$P(A | B)$ = Probability of “the event A given that B has occurred”.

We define

$$P(A | B) = \frac{P(AB)}{P(B)}, \quad (1-23)$$

provided $P(B) \neq 0$. As we show below, the above definition satisfies all probability axioms discussed earlier.

We have

$$(i) \quad P(A | B) = \frac{P(AB) \geq 0}{P(B) > 0} \geq 0, \quad (1-24)$$

$$(ii) \quad P(\Omega | B) = \frac{P(\Omega B)}{P(B)} = \frac{P(B)}{P(B)} = 1, \quad \text{since } \Omega B = B. \quad (1-25)$$

(iii) Suppose $A \cap C = \emptyset$. Then

But $AB \cap AC = \emptyset$, hence $P(AB \cup CB) = P(AB) + P(CB)$.

$$P(A \cup C | B) = \frac{P((A \cup C) \cap B)}{P(B)} = \frac{P(AB \cup CB)}{P(B)}. \quad (1-26)$$

$$P(A \cup C | B) = \frac{P(AB)}{P(B)} + \frac{P(CB)}{P(B)} = P(A | B) + P(C | B), \quad (1-27)$$

satisfying all probability axioms in (1-12). Thus (1-23) defines a legitimate probability measure.

Properties of Conditional Probability:

a. If $B \subset A$, $AB = B$, and

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{P(B)}{P(B)} = 1 \quad (1-28)$$

since if $B \subset A$, then occurrence of B implies automatic occurrence of the event A . As an example, but

$A = \{\text{outcome is even}\}$, $B = \{\text{outcome is 2}\}$,

in a dice tossing experiment. Then $B \subset A$, and $P(A | B) = 1$.

b. If $A \subset B$, $AB = A$, and

$$P(A | B) = \frac{P(AB)}{P(B)} = \frac{P(A)}{P(B)} > P(A). \quad (1-41)$$

(In a dice experiment, $A = \{\text{outcome is 2}\}$, $B = \{\text{outcome is even}\}$, so that $A \subset B$. The statement that B has occurred (outcome is even) makes the odds for “outcome is 2” greater than without that information).

c. We can use the conditional probability to express the probability of a complicated event in terms of “simpler” related events.

Total Probability Theorem

Let $A_1, A_2, \dots, A_n$ are pair wise disjoint and their union is Ω . Thus and $A_i A_j = \phi$,

$$\text{Thus } \bigcup_{i=1}^n A_i = \Omega. \quad (1-29)$$

$$B = B(A_1 \cup A_2 \cup \dots \cup A_n) = BA_1 \cup BA_2 \cup \dots \cup BA_n. \quad (1-30)$$

But $A_i \cap A_j = \phi \Rightarrow BA_i \cap BA_j = \phi$, so that from (1-43)

$$P(B) = \sum_{i=1}^n P(BA_i) = \sum_{i=1}^n P(B | A_i)P(A_i). \quad (1-31)$$

With the notion of conditional probability, next we introduce the notion of “independence” of events.

Independence: A and B are said to be independent events, if

$$P(AB) = P(A) \cdot P(B). \quad (1-32)$$

Notice that the above definition is a probabilistic statement, *not* a set theoretic notion such as mutually exclusiveness.

Suppose A and B are independent, then

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A). \quad (1-33)$$

Thus if A and B are independent, the event that B has occurred does not shed any more light into the event A . It makes no difference to A whether B has occurred or not. An example will clarify the situation:

Example 1.2: A box contains 6 white and 4 black balls. Remove two balls at random without replacement. What is the probability that the first one is white and the second one is black?

Let W_1 = “first ball removed is white”

B_2 = “second ball removed is black”

We need $P(W_1 \cap B_2)$. We have $W_1 \cap B_2 = W_1 B_2 = B_2 W_1$. Using the conditional probability rule,

$$P(W_1 B_2) = P(B_2 W_1) = P(B_2 | W_1) P(W_1). \quad (1-34)$$

But

$$P(W_1) = \frac{6}{6+4} = \frac{6}{10} = \frac{3}{5},$$

and

$$P(B_2 | W_1) = \frac{4}{5+4} = \frac{4}{9},$$

and hence

$$P(W_1 B_2) = \frac{3}{5} \cdot \frac{4}{9} = \frac{12}{45} \approx 0.27.$$

Are the events W_1 and B_2 independent? Our common sense says No. To verify this, we need to compute $P(B_2)$. Of course, the fate of the second ball very much depends on that of the first ball. The first ball has two options:

W_1 = “first ball is white” or B_1 = “first ball is black”. Note that $W_1 \cap B_1 = \phi$,
and $W_1 \cup B_1 = \Omega$. Hence W_1 together with B_1 form a partition. Thus (see (1-23)-(1-25))

$$\begin{aligned} P(B_2) &= P(B_2 | W_1)P(W_1) + P(B_2 | B_1)P(B_1) \\ &= \frac{4}{5+4} \cdot \frac{3}{5} + \frac{3}{6+3} \cdot \frac{4}{10} = \frac{4}{9} \cdot \frac{3}{5} + \frac{1}{3} \cdot \frac{2}{5} = \frac{4+2}{15} = \frac{2}{5}, \end{aligned}$$

and

$$P(B_2)P(W_1) = \frac{2}{5} \cdot \frac{3}{5} \neq P(B_2 W_1) = \frac{20}{81}.$$

As expected, the events W_1 and B_2 are dependent.

Baye's Theorem

From (1-22), $P(AB) = P(A | B)P(B).$ (1-35)

Similarly, from (1-22) $P(B | A) = \frac{P(BA)}{P(A)} = \frac{P(AB)}{P(A)},$

or $P(AB) = P(B | A)P(A).$ (1-36)

From (1-35)-(1-36), we get

$$P(A | B)P(B) = P(B | A)P(A).$$

or $P(A | B) = \frac{P(B | A)}{P(B)} \cdot P(A)$ (1-37)

Equation (1-37) is known as Bayes' theorem.

A more general version of Bayes' theorem involves partition of Ω . From (1-37)

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{P(B)} = \frac{P(B | A_i)P(A_i)}{\sum_{i=1}^n P(B | A_i)P(A_i)}, \quad (1-38)$$

where we have made use of (1-31). In (1-38), $A_i, i = 1 \rightarrow n$, represent a set of mutually exclusive events with associated a-priori probabilities

$P(A_i), i = 1 \rightarrow n$. With the new information “ B has occurred”, the information about A_i can be updated by the n conditional probabilities

$$P(B | A_i), i = 1 \rightarrow n, \text{ using (35)}).$$

Example 1.3: Two boxes B_1 and B_2 contain 100 and 200 light bulbs respectively. The first box (B_1) has 15 defective bulbs and the second 5. Suppose a box is selected at random and one bulb is picked out.

(a) What is the probability that it is defective?

Solution: Note that box B_1 has 85 good and 15 defective bulbs. Similarly box B_2 has 195 good and 5 defective bulbs. Let D = “Defective bulb is picked out”.

Then

$$P(D \mid B_1) = \frac{15}{100} = 0.15, \quad P(D \mid B_2) = \frac{5}{200} = 0.025.$$

Since a box is selected at random, they are equally likely.

$$P(B_1) = P(B_2) = \frac{1}{2}.$$

Thus B_1 and B_2 form a partition as in (1-30), and using (1-31) we obtain

$$\begin{aligned} P(D) &= P(D | B_1)P(B_1) + P(D | B_2)P(B_2) \\ &= 0.15 \times \frac{1}{2} + 0.025 \times \frac{1}{2} = 0.0875. \end{aligned}$$

Thus, there is about 9% probability that a bulb picked at random is defective.

(b) Suppose we test the bulb and it is found to be defective. What is the probability that it came from box 1? $P(B_1 | D) = ?$

$$P(B_1 | D) = \frac{P(D | B_1)P(B_1)}{P(D)} = \frac{0.15 \times 1/2}{0.0875} = 0.8571. \quad (1-39)$$

Notice that initially $P(B_1) = 0.5$; then we picked out a box at random and tested a bulb that turned out to be defective. Can this information shed some light about the fact that we might have picked up box 1?

From (1-39), $P(B_1 | D) = 0.857 > 0.5$, and indeed it is more likely at this point that we must have chosen box 1 in favor of box 2. (Recall box1 has six times more defective bulbs compared to box2).

RANDOM VARIABLE

- A random variable x takes on a defined set of values with different probabilities.
 - For example, if you roll a die, the outcome is random (not fixed) and there are 6 possible outcomes, each of which occur with probability one-sixth.
 - For example, if you poll people about their voting preferences, the percentage of the sample that responds “Yes on Proposition 100” is also a random variable (the percentage will be slightly differently every time you poll).

NOTATION

Random Variables are generally denoted with capital letters such as X , Y , or Z .

The letter Z is often reserved for random variables that follow a standardized **normal** distribution.

EXAMPLE: *A SIMPLE RANDOM VARIABLE*

- | Toss a coin twice, and let the random variable X be the number of tails appearing.
- | **Question:**
 - | What are the possible values of X ?
 - | Possible values: 0, 1, 2

DEFINITIONS

- | **Probability distribution** of a random variable tells all of its possible values along with their associated probabilities.
- | **Probability histogram** displays possible values of a random variable along horizontal axis, probabilities along vertical axis.

EXAMPLE: PROBABILITY DISTRIBUTION OF A RANDOM VARIABLE

The random variable X is the number of tails in two tosses of a coin.

Questions:

What are the probabilities of the possible outcomes?

What is the probability distribution of X ?

Possible outcomes:

Each has probability $\frac{1}{4}$ so the probability distribution is:



Non-overlapping “Or” Rule $\square P(X=1)=1/2$

X = Number of tails	0	1	2
Probability	$1/4$	$1/2$	$1/4$

EXAMPLE: *PERMISSIBLE PROBABILITIES RULE*

X	1	2	3	4	5	6	7
$P(X = x)$	0.26	0.34	0.16	0.14	0.07	0.02	0.01

Household size in U.S. has

Question: How do these probabilities conform to the [Permissible Probabilities Rule](#)?

Answer: All between 0 and 1.

EXAMPLE: INTERIM TABLE AND PROBABILITY DISTRIBUTION

A coin is tossed 3 times and the random variable X is number of tails tossed.

Questions: What are the possible outcomes, values of X , and probabilities?
How do we find probability that $X=1$? $X=2$?

Interim Table:

Use Non-overlapping “Or” Rule to combine probabilities

Outcome	$X=\text{no. of tails}$	Probability
HHH	0	$1/8$
HHT	1	$1/8$
HTH	1	$1/8$
THH	1	$1/8$
HTT	2	$1/8$
THT	2	$1/8$
TTH	2	$1/8$
TTT	3	$1/8$

RANDOM VARIABLES CAN BE DISCRETE OR CONTINUOUS

- | **Discrete** random variables have a countable number of outcomes
 - | Examples: Dead/alive, treatment/placebo, dice, counts, etc.
- | **Continuous** random variables have an infinite continuum of possible values.
 - | Examples: blood pressure, weight, the speed of a car, the real numbers from 1 to 6.

PROBABILITY FUNCTIONS

- A probability function maps the possible values of x against their respective probabilities of occurrence, $p(x)$
- $p(x)$ is a number from 0 to 1.0.
- The area under a probability function is always 1.

Continuous Random Variables

Probability Density Function

- The **probability distribution** or simply **distribution** of a random variable X is a description of the set of the probabilities associated with the possible values for X .

The **probability density function** (or pdf) $f(x)$ of a continuous random variable is used to determine probabilities from areas as follows:

$$P(a < X < b) = \int_a^b f(x) dx \quad (3-2)$$

The properties of the pdf are

(1) $f(x) \geq 0$

(2) $\int_{-\infty}^{\infty} f(x) = 1$

Continuous Random Variables

Probability Density Function

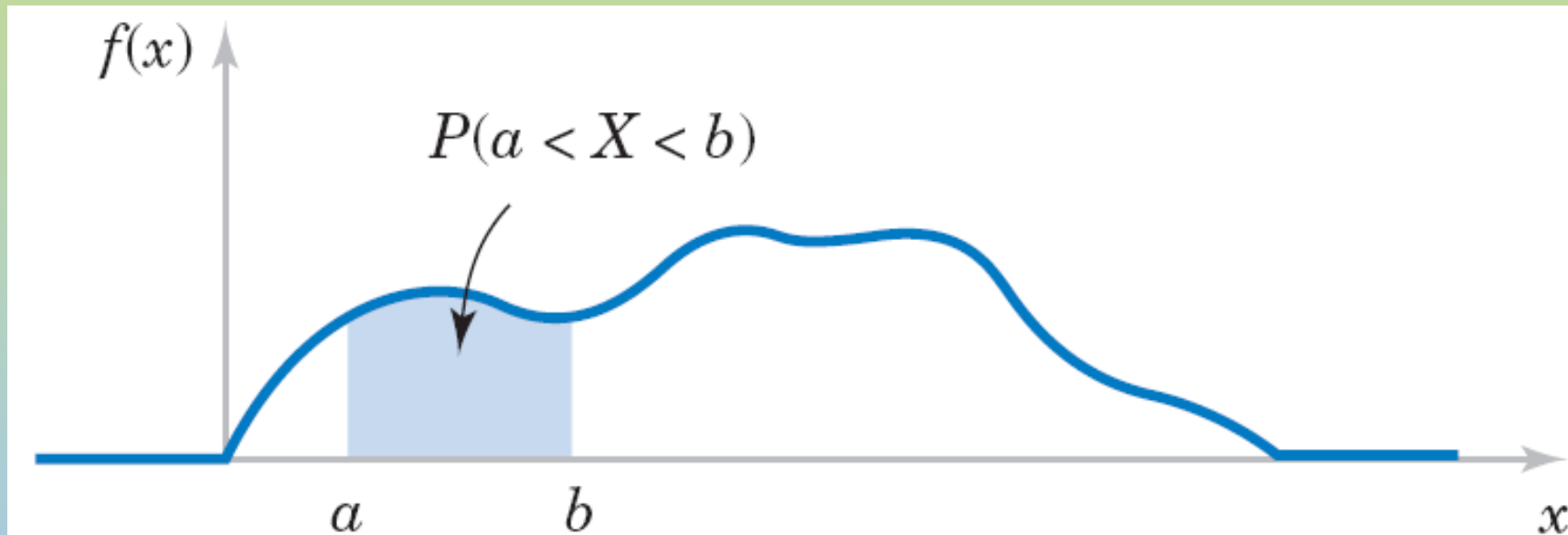


Figure . Probability determined from the area under $f(x)$.

Continuous Random Variables

Probability Density Function

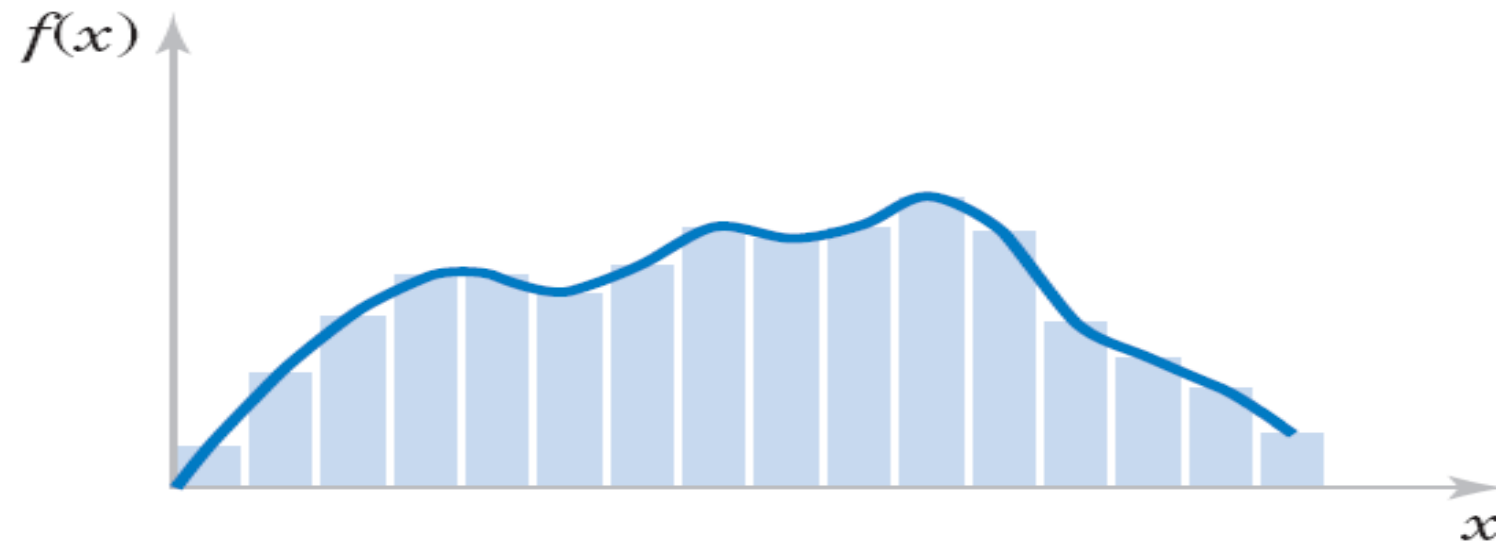


Figure A histogram approximates a probability density function. The area of each bar equals the relative frequency of the interval. The area under $f(x)$ over any interval equals the probability of the interval.

Continuous Random Variables

Probability Density Function

If X is a continuous random variable, for any x_1 and x_2 ,

$$P(x_1 \leq X \leq x_2) = P(x_1 < X \leq x_2) = P(x_1 \leq X < x_2) = P(x_1 < X < x_2)$$

Continuous Random Variables

Let the continuous random variable X denote the distance in micrometers from the start of a track on a magnetic disk until the first flaw. Historical data show that the distribution of X can be modeled by a pdf $f(x) = \frac{1}{2000} e^{-x/2000}$, $x \geq 0$. For what proportion of disks is the distance to the first flaw greater than 1000 micrometers?

Solution. The density function and the requested probability are shown in Fig. 3-9. Now,

$$P(X > 1000) = \int_{1000}^{\infty} f(x) dx = \int_{1000}^{\infty} \frac{e^{-x/2000}}{2000} dx = -e^{-x/2000} \Big|_{1000}^{\infty} = e^{-1/2} = 0.607$$

What proportion of parts is between 1000 and 2000 micrometers?

Solution. Now,

$$P(1000 < X < 2000) = \int_{1000}^{2000} f(x) dx = -e^{-x/2000} \Big|_{1000}^{2000} = e^{-1/2} - e^{-1} = 0.239$$

Because the total area under $f(x)$ equals 1, we can also calculate $P(X < 1000) = 1 - P(X > 1000) = 1 - 0.607 = 0.393$. 

Continuous Random Variables

Cumulative Distribution Function

The **cumulative distribution function** (or cdf) of a continuous random variable X with probability density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du$$

for $-\infty < x < \infty$.

Discrete Random Variables

- Only measurements at **discrete** points are possible

Discrete Random Variables

Probability Mass Function

For a discrete random variable X with possible values $x_1, x_2, \dots, x_n$, the **probability mass function** (or pmf) is

$$f(x_i) = P(X = x_i)$$

PMF

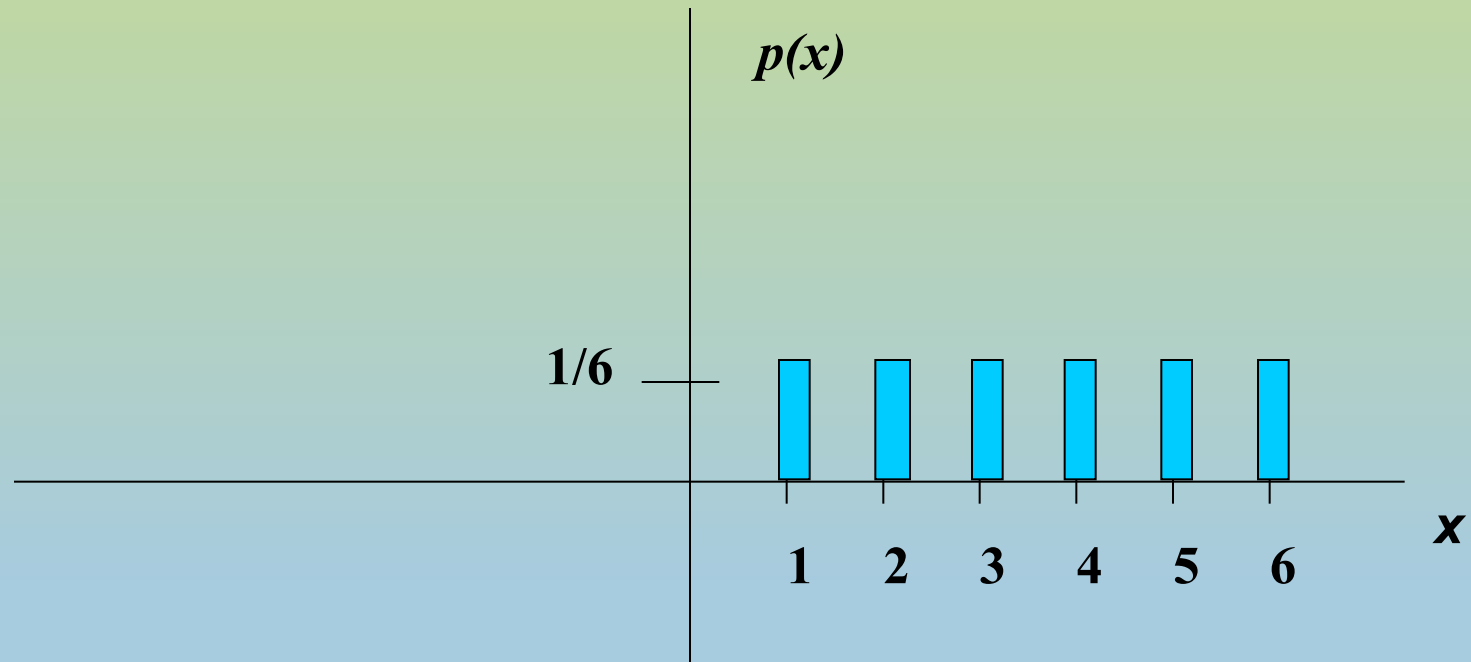
Discrete Random Variables

Cumulative Distribution Function

The **cumulative distribution function** of a discrete random variable X is

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

DISCRETE EXAMPLE: ROLL OF A DIE



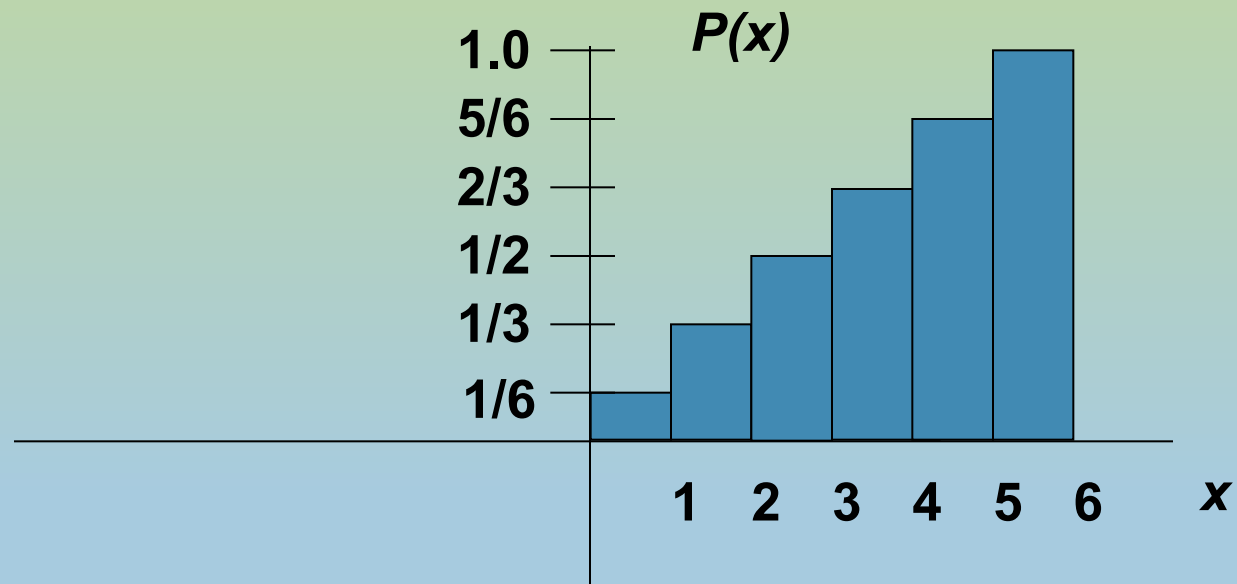
$$\sum_{\text{all } x} P(x) = 1$$

PROBABILITY MASS FUNCTION (PMF)

x	$p(x)$
1	$p(x=1)=1/6$
2	$p(x=2)=1/6$
3	$p(x=3)=1/6$
4	$p(x=4)=1/6$
5	$p(x=5)=1/6$
6	<u>$p(x=6)=1/6$</u>

1.0

CUMULATIVE DISTRIBUTION FUNCTION (CDF)



CUMULATIVE DISTRIBUTION FUNCTION

x	$P(x \leq A)$
1	$P(x \leq 1) = 1/6$
2	$P(x \leq 2) = 2/6$
3	$P(x \leq 3) = 3/6$
4	$P(x \leq 4) = 4/6$
5	$P(x \leq 5) = 5/6$
6	$P(x \leq 6) = 6/6$

PRACTICE PROBLEM

Which of the following are probability functions?

- a. $f(x) = .25$ for $x = 9, 10, 11, 12$
- b. $f(x) = (3-x)/2$ for $x = 1, 2, 3, 4$
- c. $f(x) = (x^2 + x + 1)/25$ for $x = 0, 1, 2, 3$

ANSWER (A)

- a. $f(x) = .25$ for $x=9,10,11,12$

x	$f(x)$
9	.25
10	.25
11	.25
12	<u>.25</u>
	1.0

Yes, probability
function!

PRACTICE PROBLEM:

x	10	11	12	13	14
$P(x)$	.4	.2	.2	.1	.1

- | The number of ships to arrive at a harbor on any given day is a random variable represented by x . The probability distribution for x is:

Find the probability that on a given day:

- a. exactly 14 ships arrive $p(x=14) = .1$
- b. At least 12 ships arrive $p(x \geq 12) = (.2 + .1 + .1) = .4$
- c. At most 11 ships arrive $p(x \leq 11) = (.4 + .2) = .6$

EXPECTED VALUE AND VARIANCE

- | All probability distributions are characterized by an expected value and a variance (standard deviation squared).

EXPECTED VALUE, OR MEAN

- | If we understand the underlying probability function of a certain phenomenon, then we can make informed decisions based on how we expect x to behave on-average over the long-run...(so called “frequentist” theory of probability).
- | Expected value is just the weighted average or mean (μ) of random variable x . Imagine placing the masses $p(x)$ at the points X on a beam; the balance point of the beam is the expected value of x .

Discrete Random Variables

Mean and Variance

Let the possible values of the random variable X be denoted as $x_1, x_2, \dots, x_n$. The pmf of X is $f(x)$, so $f(x_i) = P(X = x_i)$.

The **mean** or **expected value** of the discrete random variable X , denoted as μ or $E(X)$, is

$$\mu = E(X) = \sum_{i=1}^n x_i f(x_i)$$

The **variance** of X , denoted as σ^2 or $V(X)$, is

$$\sigma^2 = V(X) = E(X - \mu)^2 = \sum_{i=1}^n (x_i - \mu)^2 f(x_i) = \sum_{i=1}^n x_i^2 f(x_i) - \mu^2$$

The **standard deviation** of X is σ .

EXPECTED VALUE, FORMALLY

Discrete case:

$$E(X) = \sum_{\text{all } x} x_i p(x_i)$$

Continuous case:

$$E(X) = \int_{\text{all } x} x_i p(x_i) dx$$

EMPIRICAL MEAN IS A SPECIAL CASE OF EXPECTED VALUE...

Sample mean, for a sample of n subjects: =

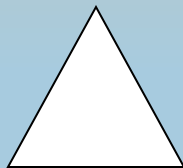
$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n} = \sum_{i=1}^n x_i \left(\frac{1}{n} \right)$$


The probability (frequency) of each person in the sample is $1/n$.

EXAMPLE: EXPECTED VALUE

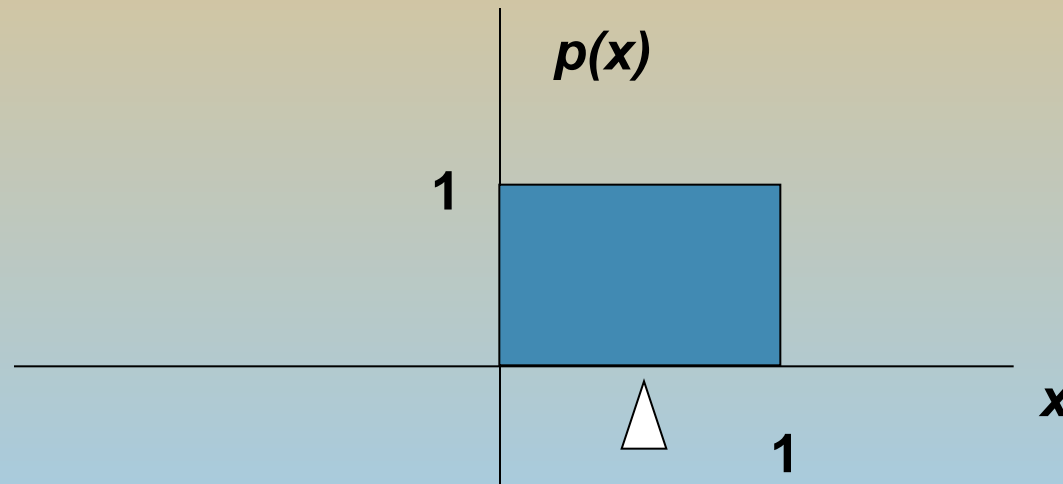
x	10	11	12	13	14
$P(x)$	.4	.2	.2	.1	.1

Recall the following probability distribution of ship arrivals:



$$\sum_{i=1}^5 x_i p(x) = 10(.4) + 11(.2) + 12(.2) + 13(.1) + 14(.1) = 11.3$$

EXTENSION TO CONTINUOUS CASE: UNIFORM DISTRIBUTION



$$E(X) = \int_0^1 x(1)dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2} - 0 = \frac{1}{2}$$

**A FEW NOTES ABOUT EXPECTED VALUE AS A MATHEMATICAL OPERATOR:

If c = a constant number (i.e., not a variable) and X and Y are any random variables...

↑ $E(c) = c$

↑ $E(cX) = cE(X)$

↑ $E(c + X) = c + E(X)$

↑ $E(X + Y) = E(X) + E(Y)$

VARIANCE/STANDARD DEVIATION

$$\sigma^2 = Var(x) = E[(x - \mu)^2] = \sum_{\text{all } x} (x_i - \mu)^2 p(x_i)$$

“The average (expected) squared distance (or deviation) from the mean”

***We square because squaring has better properties than absolute value. Take square root to get back linear average distance from the mean (=“standard deviation”).*

VARIANCE, FORMALLY

Discrete case:

$$Var(X) = \sigma^2 = \sum_{\text{all } x} (x_i - \mu)^2 p(x_i)$$

Continuous case:

$$Var(X) = \sigma^2 = \int_{-\infty}^{\infty} (x_i - \mu)^2 p(x_i) dx$$

SIMILARITY TO EMPIRICAL VARIANCE

The variance of a sample: $s^2 =$

$$\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{n-1} = \sum_{i=1}^N (x_i - \bar{x})^2 \left(\frac{1}{n-1} \right)$$

Division by $n-1$ reflects the fact that we have lost a “degree of freedom” (piece of information) because we had to estimate the sample mean before we could estimate the sample variance.


$$\text{VAR}(C) = 0$$

$$\text{Var}(c) = 0$$

Constants don't vary!

PRACTICE PROBLEM

Find the variance and standard deviation for the number of ships to arrive at the harbor (recall that the mean is 11.3).

x	10	11	12	13	14
$P(x)$	.4	.2	.2	.1	.1

ANSWER: VARIANCE AND STD DEV

x^2	100	121	144	169	196
$P(x)$	.4	.2	.2	.1	.1

$$E(x^2) = \sum_{i=1}^5 x_i^2 p(x_i) = (100)(.4) + (121)(.2) + 144(.2) + 169(.1) + 196(.1) = 129.5$$

$$Var(x) = E(x^2) - [E(x)]^2 = 129.5 - 11.3^2 = 1.81$$

$$stddev(x) = \sqrt{1.81} = 1.35$$

Interpretation: On an average day, we expect 11.3 ships to arrive in the harbor, plus or minus 1.35. This gives you a feel for what would be considered a usual day!



PRACTICE PROBLEM

You toss a coin 100 times. What's the expected number of heads? What's the variance of the number of heads?

ANSWER: EXPECTED VALUE

Intuitively, we'd probably all agree that we expect around 50 heads, right?

Another way to show this $\square$

Think of tossing 1 coin. $E(X=\text{number of heads}) = (1)P(\text{heads}) + (0)P(\text{tails})$

$$\square E(X=\text{number of heads}) = 1(.5) + 0 = .5$$

If we do this 100 times, we're looking for the sum of 100 tosses, where we assign 1 for a heads and 0 for a tails. (these are 100 "independent, identically distributed (i.i.d)" events)

$$E(X_1 + X_2 + X_3 + X_4 + X_5 + \dots + X_{100}) = E(X_1) + E(X_2) + E(X_3) + E(X_4) + E(X_5) + \dots + E(X_{100}) = 100 E(X_1) = 50$$

ANSWER: VARIANCE

What's the variability, though? More tricky. But, again, we could do this for 1 coin and then use our rules of variance.

Think of tossing 1 coin.

$$E(X^2 = \text{number of heads squared}) = 1^2 P(\text{heads}) + 0^2 P(\text{tails})$$

$$\square E(X^2) = 1(.5) + 0 = .5$$

$$\text{Var}(X) = .5 - .5^2 = .5 - .25 = .25$$

Then, using our rule: $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$ (coin tosses are independent!)

$$\text{Var}(X_1 + X_2 + X_3 + X_4 + X_5 + \dots + X_{100}) = \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) + \text{Var}(X_4) + \text{Var}(X_5) + \dots + \text{Var}(X_{100}) =$$

$$100 \text{Var}(X_1) = 100 (.25) = 25$$

$$\text{SD}(X) = 5$$

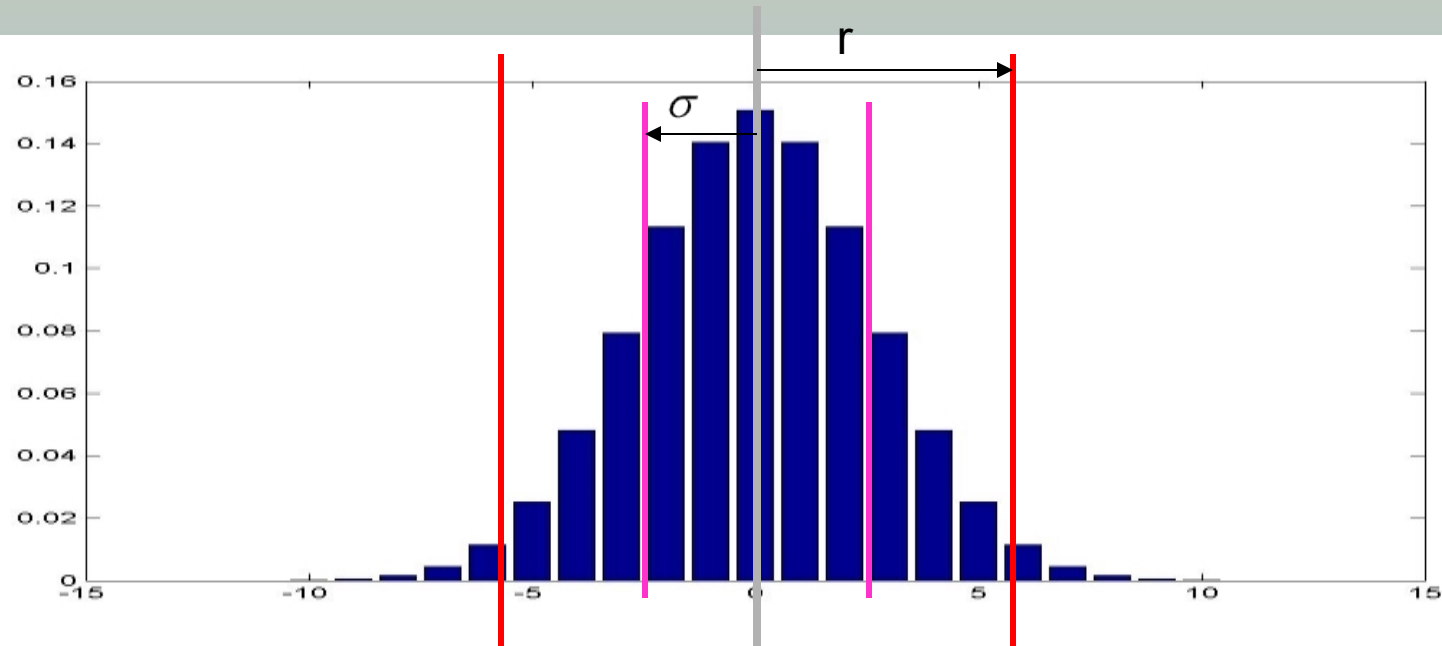
Interpretation: When we toss a coin 100 times, we expect to get 50 heads plus or minus 5.

CHEBYSHEV'S INEQUALITY

Theorem: *X is random variable on sample space S , and $P(X=r)$ it's probability distribution. Then for any positive real number r :*

$$P(|X - E(X)| \geq r) \leq \frac{V(X)}{r^2}$$

In words: the probability of finding a value of X farther away from the mean than r is smaller than the variance divided by r^2 .



CHEBYSHEV THEOREM

How to calculate the probability when the mean value is known?

Use the Chebyshev Theorem:

X is non-negative with a finite expected value;
 k is a positive constant

$$\Pr(X \geq k) \leq \frac{E(X)}{k}$$

CHEBYSHEV THEOREM

Question:

What is the probability that X falls into an interval:

$$\mu \pm k\sigma$$

re-write the Chebyshev inequality

k – non-negative constant, m – mean value, s – standard deviation

CHEBYSHEV THEOREM

Probabilities:

$$P\left[|X - \mu| < k\sigma\right] \geq 1 - \frac{1}{k^2}$$

$$P\left[|X - \mu| \geq k\sigma\right] \leq \frac{1}{k^2}$$



k – non-negative constant, m – mean value, s – standard deviation

CHEBYSHEV THEOREM

How to calculate the k :

$$k = \frac{X - \mu}{\sigma}$$

How to choose X :

arbitrary choice

upper limit (*maximum value*)

aim (*planned value*)

k – non-negative constant, m – mean value, s – standard deviation

CHEBYSHEV THEOREM

Another method of selecting k :
compare the difference between X and the
mean value with a number (e.g. z):

$$X - \mu < k\sigma = z$$

$$z = k\sigma$$

$$k = \frac{z}{\sigma}$$

k – non-negative constant, m – mean value, s – standard deviation

CHEBYSHEV THEOREM

Data:

avg. production = 120 units

std. deviation = 10 units

Problem:

what is the probability that production will fall
within 100-140 unit interval

$$k = (140-120)/10; k = (120-100)/10$$

Example: What is the probability that with 100 Bernoulli trials we find more than 89 or less than 11 successes when the prob. of success is $\frac{1}{2}$.

X counts number of successes.

$$EX = 100 \times \frac{1}{2} = 50$$

$$V(X) = 100 \times \frac{1}{2} \times \frac{1}{2} = 25.$$

$$P(|X-50| \geq 40) \leq \frac{25}{40^2} = \frac{1}{64}$$

Let $f(x) = \frac{5}{x^6}$ for $x \geq 1$ and 0 otherwise. What bound does Chebyshev's inequality give for the probability $P(X \geq 2.5)$? For what value of a can we say $P(X \geq a) \leq 15\%$?

Solution: We calculate the mean as

$$\mu = \int_{-\infty}^{\infty} x f(x) dx = \int_1^{\infty} \frac{5}{x^5} dx = \frac{5}{4}.$$

The variance is

$$\sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2 = \int_1^{\infty} \frac{5}{x^4} dx - \frac{25}{16} = \frac{5}{3} - \frac{25}{16} = \frac{5}{48}.$$

In order to use Chebyshev's, we need to have a symmetric probability. Namely, we can only use it for something like $P(|X - \mu| \geq r)$. We write $P(X \geq 2.5) = P(X \geq \frac{5}{4} + \frac{5}{4})$. So write

$$P(|X - \mu| \geq \frac{5}{4}) = P(X \geq 2.5 \text{ or } X \leq 0) = P(X \geq 2.5) \leq \frac{\sigma^2}{(5/4)^2} = \frac{5/48}{25/16} = \frac{1}{15}.$$

If we want $P(X \geq a) \leq 15\% = 0.15 = \frac{15}{100}$ using Chebyshev's inequality, then we need

$$\frac{15}{100} = \frac{\sigma^2}{r^2} = \frac{5/48}{r^2}$$

So

$$r = \sqrt{\frac{5/48}{15/100}} = \sqrt{\frac{100}{3 \cdot 48}} = \frac{10}{12} = \frac{5}{6}.$$

Therefore, we need $P(X \geq \mu + 5/6)$ and $a = \frac{5}{4} + \frac{5}{6} = \frac{25}{12}$.